Robust Face Detection Using Template Matching Algorithm

by

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A thesis submitted in conformity with the requirements
for the degree of Masters of Applied Science
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Abstract

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Human face detection and recognition techniques play an important role in applications like face recognition, video surveillance, human computer interface and face image databases. Using color information in images is one of the various possible techniques used for face detection. The novel technique used in this project was the combination of various techniques such as skin color detection, template matching, gradient face detection to achieve high accuracy of face detection in frontal faces. The objective in this work was to determine the best rotation angle to achieve optimal detection. Also eye and mouse template matching have been put to test for feature detection.

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Chapter 1

Introduction

During the past two decades, both consumer and business worlds have witnessed a rapid growth in video and image processing to fulfill the needs of object detection for various applications such as data query and object retrievals. One of the most widely researched areas, thoroughly investigated for various applications is face object. This chapter briefly discusses the rationale behind the development of face detectors and subsets of these systems.

1.1 Motivation

Traditionally, computer vision systems have been used in specific tasks, such as performing tedious and repetitive visual tasks of assembly line inspection. Current development in this area has moved toward more generalized vision applications such as face recognition, video coding techniques, biometrics, surveillance, man-machine interaction, animation and database indexing and many other applications that have face detection as the primary building block of their systems.

Many of the current face recognition systems assume the availability of frontal faces. In reality this assumption may not hold due to the nature of face appearance and environment conditions. The exclusion of background in these images is necessary for reliable

face classification. However in realistic application scenarios a face could occur in a complex background and in many different positions. Recognition systems that are built on the standard face images are likely to mistake areas of the background as faces. In order to rectify the problem, a visual processor is needed to localize and extract the face region from the background.

Face detection is one of the visual tasks that humans can do effortlessly. However, in computer vision terms, this task is not easy. A general statement of the problem can be defined as follows: Given an image or a video sequence, detect and localize an unknown number(if any) of faces. The solution to this problem involves segmentation, extraction and verification of faces and possibly facial features from an uncontrolled background. An ideal face detector should achieve this aim despite illumination, rotation, different facial expressions, orientations and camera distance from the object.

In the last two decades huge progress has been made to increase the accuracy of the face detectors while many different methods have been introduced in this area. A short survey on different methods will be introduced in the next section.

1.2 Face Detection Methods

Research on face detection started in the beginning of the 1970s, where simple heuristic and anthropometric techniques [16, 18] were used. The techniques that were used were too rigid and worked only on the plain background and any challenge would confuse the system to perform properly. Despite these problems the growth of research interest remained stagnant until the 1990s [15, 18], when practical face recognition and video coding systems started to become a reality.

Over the past two decades there has been a great deal of research interest spanning several important aspects of face detection. More robust segmentation schemes have been presented, particularly those using motion, color, and generalized information. The use of

statistics and neural networks has also enabled faces to be detected from cluttered scenes at different distances from the camera. Additionally, there are numerous advances in the design of feature extractors such as the deformable templates and the active contours which can locate and track facial features accurately. Because face detection techniques require a priori information of the face, they can be effectively organized into two broad categories distinguished by their different approach to utilizing face knowledge. The techniques in the first category make explicit use of face knowledge and follow the classical detection methodology in which low level features are derived prior to knowledge-based analysis [17, 18].

The apparent properties of the face such as skin color and face geometry are exploited at different system levels. Typically, in these techniques face detection tasks are accomplished by manipulating distance, angles, and area measurements of the visual features derived from the scene. Since features are the main ingredients, these techniques are termed the feature-based approach. These approaches have embodied the majority of interest in face detection research starting as early as the 1970s and therefore account for most of the literature reviewed herein. Taking advantage of the current advances in pattern recognition theory, the techniques in the second group address face detection as a general recognition problem. Image-based [18] representations of faces, for example in 2D intensity arrays, are directly classified into a face group using training algorithms without feature derivation and analysis. Unlike the feature-based approach, these relatively new techniques incorporate face knowledge implicitly [17, 18] into the system through mapping and training schemes. Different methods that were just mentioned in this section are shown in figure 1.1.

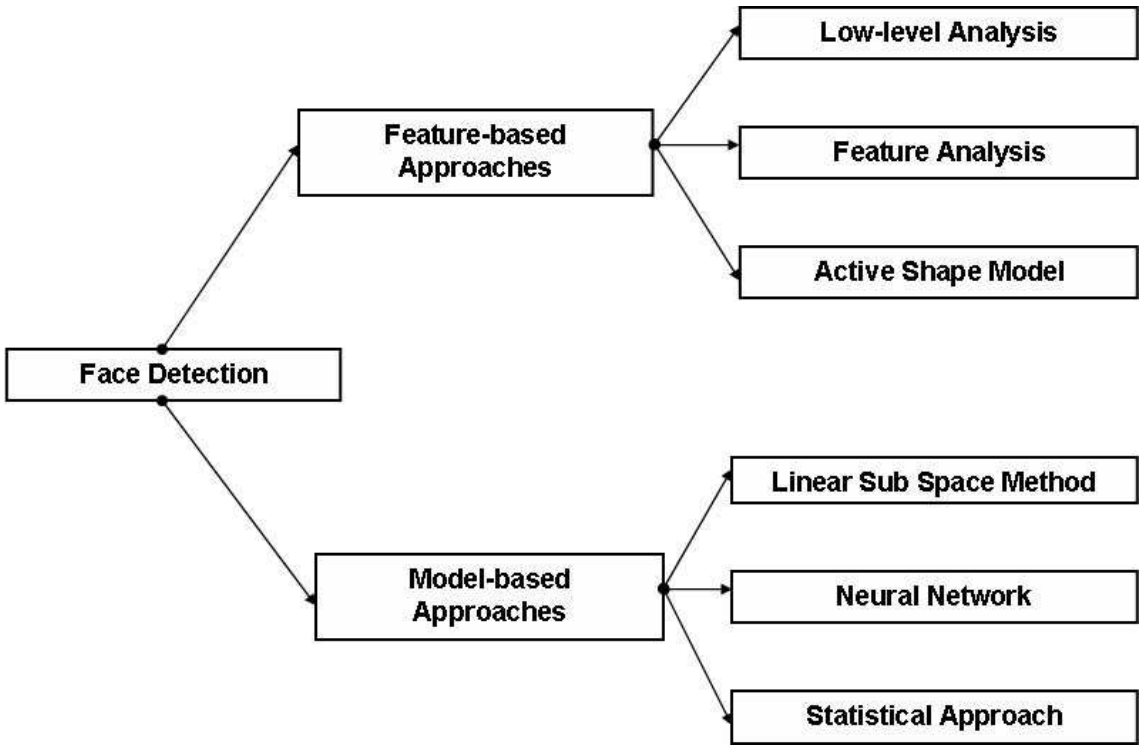

Figure 1.1: Different Approaches To Face Detection

1.3 Feature-Based Approaches

The development of feature based approach can be further divided into three sub categories as shown in figure 1.1. Low level analysis first dealt with segmentation of visual features using pixel properties such as edge detection, gray scale analysis, colour information. Features generated from low-level analysis are likely to be ambiguous. For instance, in locating facial regions using a skin colour model, background objects of similar colour can also be detected. This is a classical many to one mapping problem which can be solved by higher level feature analysis. In many face detection techniques, the knowledge of face geometry has been employed to characterize and subsequently verify various features from their ambiguous state. There are two approaches in the application of face geometry among the literature surveyed. The first approach involves sequential feature searching strategies based on the relative positioning of individual facial features. The confidence of a feature is enhanced by the detection of nearby features. The techniques in the second approach group feature as flexible constellations using various face models.

1.3.1 Feature searching

Feature searching techniques begin with the determination of prominent facial features. The detection of the prominent features then allows for the existence of other less prominent features to be hypothesized using anthropometric measurements of face geometry. Among the literature survey, a pair of eyes is the most commonly applied reference feature [18–21] due to its distinct side-by-side appearance. Other features include a main face axis, outline (top of the head) and body (below the head). The facial feature extraction algorithm by De R. Hsu and M. Abdel-Mottaleb [22] is a good example of feature searching. The algorithm starts by finding different face features such as eyes and mouth and score the face candidates accordingly. APL lab under supervision of P. Aarabi has also been working on such projects for the last 10 years [38–40]. Jeng et al. [23] pro-

pose a system for face and facial feature detection which is also based on anthropometric measures. In their system, they initially try to establish possible locations of the eyes in binarized pre-processed images. For each possible eye pair the algorithm goes on to search for a nose, a mouth, and eyebrows. Each facial feature has an associated evaluation function, which is used to determine the final most likely face candidate, weighted by their facial importance with manually selected coefficients. They report a 86 *percent* detection rate on a dataset of 114 test images taken under controlled imaging conditions, but with subjects positioned in various directions with a cluttered background.

1.3.2 Constellation analysis

Some of the algorithms mentioned in the last section rely extensively on heuristic information taken from various face images modeled under fixed conditions. If given a more general task such as locating the face(s) of various poses in complex backgrounds, many such algorithms will fail because of their rigid nature. Later efforts in face detection research address this problem by grouping facial features in face-like constellations using more robust modeling methods such as statistical analysis. Various types of face constellations have been proposed [18, 24–26]. Burl et al. [11, 12] make use of statistical shape theory on the features detected from a multi-scale Gaussian derivative filter. A probabilistic model for the spatial arrangement of facial features enables higher detection flexibility. The algorithm is able to handle missing features and problems due to translation, rotation, and scale to a certain extent and a successful rate of 84 *percent* accurate detection out of 150 images taken from a lab-scene sequence, is obtained.

1.3.3 Active Shape Model

Unlike the face models described in the previous sections, active shape models depict the actual physical and hence higher-level appearance of features. Once released within a close proximity to a feature, an active shape model will interact with local image

features (edges, brightness) and gradually deform to take the shape of the feature. There are generally three types of active shape models in the contemporary facial extraction research. The first type uses a generic active contour called snakes, first introduced by Kass et al. in 1987 [27]. Deformable templates were then introduced by Yuille et al. [29] to take into account the a priori of facial features and to better the performance of snakes. Cootes et al. [18, 30] later proposed the use of a new generic flexible model which they termed smart snakes and PDM to provide an efficient interpretation of the human face. Cootes et al.s model is based on a set of labeled points that are only allowed to vary to certain shapes according to a training procedure.

1.4 Image-Based Approach

In the previous section it was shown that the unpredictability of face appearance and environmental feature affect the accuracy of the system that only rely on explicit face features. Although some of the recent feature-based attempts have improved the ability to cope with the unpredictability, most are still limited to head and shoulder and quasi-frontal faces. There is still a need for techniques that can perform in more hostile scenarios such as detecting multiple faces with clutter-intensive backgrounds. This requirement has inspired a new research area in which face detection is treated as a pattern recognition problem. By formulating the problem as one of learning to recognize a face pattern from examples, the specific application of face knowledge is avoided. This eliminates the potential of modeling error due to incomplete or inaccurate face knowledge. The basic approach in recognizing face patterns is via a training procedure which classifies examples into face and non-face prototype classes. Comparison between these classes and a 2D intensity array (hence the name image-based) extracted from an input image allows the decision of face existence to be made [18].

Most of the image-based approaches apply a window scanning technique for detecting

faces. The window scanning algorithm is in essence just an exhaustive search of the input image for possible face locations at all scales, but there are variations in the implementation of this algorithm for almost all the image-based systems. Typically, the size of the scanning window, the sub-sampling rate, the step size, and the number of iterations vary depending on the method proposed and the need for a computationally efficient system.

Image based approach can be divided into three subsections which are as following:

1.4.1 Linear Subspace Model

Images of human faces lie in a subspace of the overall image space. To represent this subspace, one can use neural approaches but there are also several methods more closely related to standard multivariate statistical analysis which can be applied. There are many famous techniques in this category that has been used for different analysis, a few of these methods includes principal component analysis (PCA), linear discriminant analysis (LDA), and factor analysis (FA) [36, 37].

1.4.2 Neural Network

Neural networks have become a popular technique for pattern recognition problems, including face detection. Modular architectures, committee ensemble classification, complex learning algorithms, auto associative and compression networks, and networks evolved or pruned with genetic algorithms are all examples of the widespread use of neural networks in pattern recognition [18].

For face recognition, this implies that neural approaches might be applied for all parts of the system, and this had indeed been shown in several papers [32]. An introduction to some basic neural network methods for face detection can be found in Viennet and Fougelman Soulie [31].

1.4.3 Statistical Approaches

Apart from linear subspace methods and neural networks, there are several other statistical approaches to face detection. Systems based on information theory, a support vector machine and Bayes decision rule are categorized in this section.

1.5 Problem Overview

As discussed in previous sections, there are two main approaches that are taken by many researchers for face detection. In this research, feature-based technique has been studied to improve the performance and accuracy of the systems that are introduced by different researchers in previous years in the field of face detection using template matching techniques.

There are two main concerns and challenges taken in this work. One is to overcome the complexity introduced by the variety of the face angles in photos taken by arbitrary users. And second is boosting the overall accuracy of the system in comparison with the face detectors that use feature-based approach and template matching techniques as their principal solution toward face detection problem.

1.6 Outline

The rest of this thesis is organized as follows: Chapter 2 discusses the state-of-the-art developments in the area of face detection using template matching techniques; in Chapter 3 the additional features detection such as eye and mouth detector is added to the system; Also, rotation invariant characteristics are tested and the results are presented; Chapter 4 concludes this work and provides directions for future work.

Chapter 2

Prior Work

The development of the feature-based approach can be divided to three steps. Given a typical face detection problem in locating and extracting the face from the complex background, low-level analysis first deals with the segmentation of visual features using pixel properties such as colour. Due to the nature of low-level properties, face candidates that are being generated are vague. Nowadays, almost all of the videos and pictures taken by users are in colour and although the low level analysis does not pinpoint the correct answer to the face detection problem, but it sure simplifies and boosts the overall performance of the system.

Template matching algorithm, as a refiner of feature detection machine, is then developed for the purpose of complex feature extraction. Template models range from snakes, proposed in 1980s, to the more recent versions that is used in this research. The features that are of interest are the eyes and the mouth.

The last step in choosing the best face candidate is determined based on the scores associated to eyes and mouth location and symmetry. The detected face has the highest eyes and mouth likelihoods' score. Overall Face-Score determines and chooses the best face candidate among all the face candidates recognized in the previous steps.

2.1 Low-Level Analysis

2.1.1 Edge Detection

Edge detection is the primary step in deriving edge representation. Thus far, many different types of edge operators have been applied in previous works on face detection's algorithms. The Sobel operator was the most common filter used in different signal processing algorithms specially face detection [1–3].

The *Sobel* operator is a discrete differentiation operator, calculating an approximation of the gradient of the image intensity function. The operator calculates the gradient of the image intensity at each point, giving the direction of the largest possible increase from light to dark and the rate of change in that direction. The result therefore shows how sharply or smoothly the image changes at that point.

The operator uses two 3×3 kernels which are convolved with the original image to calculate approximations of the derivatives - one for vertical changes, and one for horizontal:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} \times A \tag{2.1}$$

and

$$G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \times A \tag{2.2}$$

Here G_x and G_y are the two kernels and A is the original image.

At each point in the image, the resulting gradient approximations can be combined to give the gradient magnitude, using:

$$G = \sqrt{{G_x}^2 + {G_y}^2} \tag{2.3}$$

Where G is the gradient magnitude [33].

A variety of first and second derivatives such as Laplacian filters have also been used in other methods. The Laplacian of an image $f(x, y)$, denoted by $\nabla^2 f(x, y)$, is defined as :

$$\nabla^2 f(x, y) = \frac{\partial^2 f(x, y)}{\partial x^2} + \frac{\partial^2 f(x, y)}{\partial y^2} \tag{2.4}$$

with the digital approximation, the second derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1, y) + f(x - 1, y) - 2f(x, y) \tag{2.5}$$

and

$$\frac{\partial^2 f}{\partial y^2} = f(x, y + 1) + f(x, y - 1) - 2f(x, y) \tag{2.6}$$

so that the original $\nabla^2 f(x, y)$ simplifies as following:

$$\nabla^2 f(x, y) = [f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1)] - 4f(x, y) \tag{2.7}$$

This expression can be implemented at all points (x, y) in an image by convolving the image with the following spatial mask [33]:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{2.8}$$

The edge detector formed in our face detection application used the first derivation in the vertical direction. The formulation of this derivation for an image $f(x, y)$ is as follows:

$$Gradient = \sum_{i=1}^{n-1} [f(x, y_i) - f(x, y_{i+1})] \tag{2.9}$$

where n is the number of columns in the image matrix.

2.1.2 Colour

While edge information provides the basic representation for image features, colour is a more powerful means of discerning object appearance. There are a number of colour spaces in common usage depending on the particular industry and/or application involved. For example as humans we normally determine colour by parameters such as brightness, hue, and colourfulness. On computers, one of the most widely used colour models is RGB representation in which different colours are defined by combinations of red, green, and blue primary colour components. These colours are related to the excitation of red, green, and blue phosphors on a computer monitor. Another color space widely used in different studies is $YCbCr$, where the image is being constructed by its luminance measure, namely Y, and blue and red chrominant values known as C_b and C_r respectively.

RGB Colour Space

To detect skin segments, the colour image is converted into a skin likelihood image. Since the main variation in skin appearance is largely due to luminance change (brightness), Normalized RGB colours are more preferred [4–7] The normalized RGB colour space can be calculated using the original RGB components as following:

$$r = \frac{R}{R + G + B} \tag{2.10}$$

$$g = \frac{G}{R + G + B} \tag{2.11}$$

$$b = \frac{B}{R + G + B} \tag{2.12}$$

In this approach proper threshold system, based on survey paper [8], is applied to the normalized RGB colour image to retrieve the skin areas in the image.

YCbCr Colour Space

In YC_bC_r colour model, the image is coded based on its luminance and chrominance values. Luminance characteristics are dependent on lighting conditions. On the other hand base human skin colour, though it differs widely from person to person, is distributed over a very small area on the C_bC_r (values) [28,34]. This model is robust against different types of skin, such as those of people from Europe, Asia and Africa. Based on above, the CbCr skin colour extraction formulas can be summarized as follows:

$$Y = 0.299(R - C) + C + 0.114(B - C) \tag{2.13}$$

$$Cb = 0.564(B - Y) \tag{2.14}$$

$$Cr = 0.713(R - Y) \tag{2.15}$$

2.1.3 Skin Detection

Skin colour provides good information for extracting an area of the face. The use of colour information can simplify the task of face localization in complex background [9, 10]. As is being discussed in previous section, there are many different colour spaces used in computers to represent images. Many researches have proposed various skin detection techniques based on different colour spaces [35] To detect skin patches in the images and perform segmentation on them, RGB colour spaces and YCbCr has been put to test. The segmentation procedure in the two domains are as following:

RGB Colour Space Skin Detection

The final goal of skin colour detection is to build a decision rule, that will discriminate between skin and non-skin pixels. This is usually accomplished by introducing a metric method, which measures distance (in general sense) of the pixel colour to the skin tone. The type of this metric is defined by the skin colour modeling method.

One method to build a skin classifier is to define explicitly (through a number of rules) the boundaries of the skin cluster in some colour space. The set boundaries that were well researched by P. Kovac [13], have been used for our primary skin detection.

$$
\begin{gathered}
(R, G, B) \textit{ is classified as skin if:} \\
R > 95 \textit{ and } G > 40 \textit{ and } B > 20 \textit{ and} \\
max\{R,G,B\} \text{ - } min\{R,G,B\} > 15 \textit{ and} \\
|R - G| > 15 \textit{ and } R > G \textit{ and } R > B
\end{gathered}
$$

The simplicity of this method has attracted many researchers [13, 14]. This obvious advantage, i.e. simplicity of skin detection rules, leads to construction of a very rapid classifier. The main difficulty achieving high recognition rates with this method is the need to find both appropriate colour space and adequate decision rules empirically.

The results of RGB skin detection is shown in figure 2.1.

Bayesian Skin Detection in YC_bC_r Colour Space

In YC_bC_r colour space, rather than taking the fixed boundaries, a learning machine was developed using Bayes theorem. After generating the statistics of skin color distribution, the Bayes decision rule for minimum cost [11] can be used to classify sample X into skin color class (ω_1) and non-skin color class (ω_2). Let C_ij denote the cost of deciding $X \in \omega_i$; when $X \in \omega_j$. It represents the cost of correct classification when $i = j$, and false classification when $i \neq j$. Let $R_i(X)$ denotes the conditional cost of deciding $X \in \omega_i$; given X . For the above two classes (ie: $i = 1, 2$), $R_i(X)$ is computed as

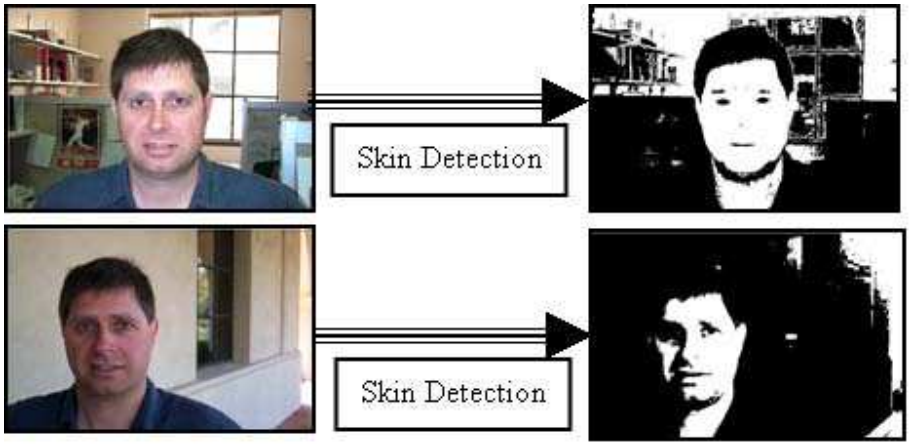

Figure 2.1: Skin Detection Results using RGB Method

$$R_1(X) = C_{11}.p(\omega_1|X) + C_{12}.p(\omega_2|X) \tag{2.16}$$

$$R_2(X) = C_{21}.p(\omega_1|X) + C_{22}.p(\omega_2|X) \tag{2.17}$$

where $p(\omega_i|X)$ denotes a posteriori probability, where the probability of being in class i is given by the sample X. The decision rule is:

$$R_1(X) \leq R_2(X) \Rightarrow X \in \omega_1 \tag{2.18}$$

$$R_1(X) \geq R_2(X) \Rightarrow X \in \omega_2 \tag{2.19}$$

The classification problem now becomes finding the class with the minimal cost. Considering different costs on the classification decisions, the previous equations can be written as following:

$$(C_{12} - C_{22}).p(\omega_2|X) \leq (C_{21} - C_{11}).p(\omega_1|x) \Rightarrow X \in \omega_1 \tag{2.20}$$

$$(C_{12} - C_{22}).p(\omega_2|X) \geq (C_{21} - C_{11}).p(\omega_1|x) \Rightarrow X \in \omega_2 \tag{2.21}$$

by applying the Bayesian Formula we achieve:

$$p(\omega_i|X) = \frac{p(X|\omega_i).p(\omega_i)}{p(X)} \tag{2.22}$$

Therefore the Bayes decision rule for minimum cost can be expressed as:

$$\frac{p(X|\omega_1)}{p(X|\omega_2)} \geq \tau \Rightarrow X \in \omega_1 \tag{2.23}$$

$$\frac{p(X|\omega_1)}{p(X|\omega_2)} \geq \tau \Rightarrow X \in \omega_1 \tag{2.24}$$

where,

$$\tau = \frac{(C_{12} - C_{22})}{(C_{21} - C_{11})}.\frac{(p(\omega_2))}{(p(\omega_1))} \tag{2.25}$$

In the above equations, $p(X|\omega_i)$ is the conditional probability density function of skin colour (when $i = 1$) and non-skin colour (when $i = 2$); $p(\omega_i)$ is the a priori probability of class ω_i; and τ represents the adjustable threshold. Note that the costs of false classifications are manipulated by C_{12} and C_{21} for false detection and false dismissal, respectively, while the costs of correct classifications(i.e. C_{11} and C_{22}) are typically set to zero [12]. The results using the above method on $YCbCr$ images are shown in figure 2.2.

2.2 High-Level Analysis

After analysing the image in low level and extracting the skin tone areas and the edges in an image, it is required to determine whether the patches are faces or not. To achieve this goal, template matching algorithm is used as the main face detector in the system.

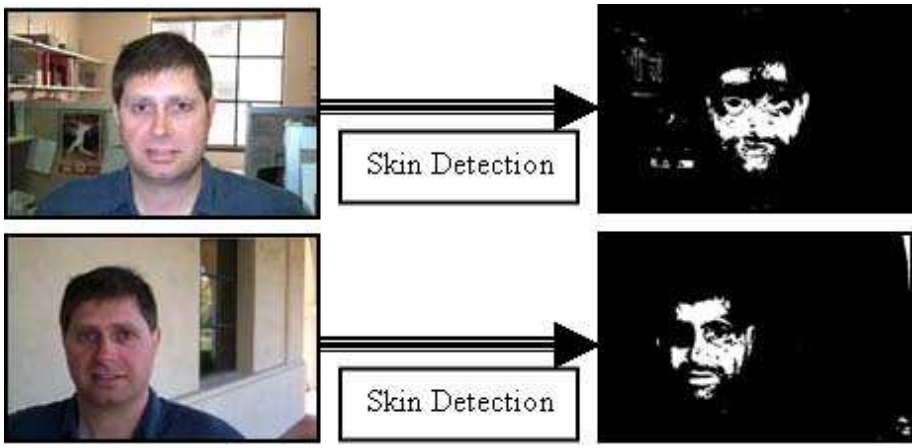

Figure 2.2: Skin Detection Results using YCbCr Method

2.2.1 Template Matching

According to a theory called Template Matching, in order to recognize an object, humans compare it to images of similar objects that they already have stored in memory. through comparing to a variety of stored candidates, it is possible to identify the object by the one that it most closely resembles. In image processing concept, a very similar idea has been used for detecting different objects in the image.

In a template matching system there is a training phase, in which a directory of image examples is processed by a digital computer to derive component vectors. As well there is a search phase, in which a digital computer processes a target image with vectors selected using component vectors to determine the presence of one or more image examples in the target image. The training phase can be conducted off line in order to come up with a template that can match the objects that are of most interest in the target image. In the searching phase that was designed in our algorithm, the template searches through the scaled binary image. The search-box runs exhaustively over the scaled down version of the original image. Each time, the template is tried to be matched over the search area if the skin patch underneath it is greater than the threshold value. Figure 2.3 shows the template that is being used for face detection purpose. In Template matching, the

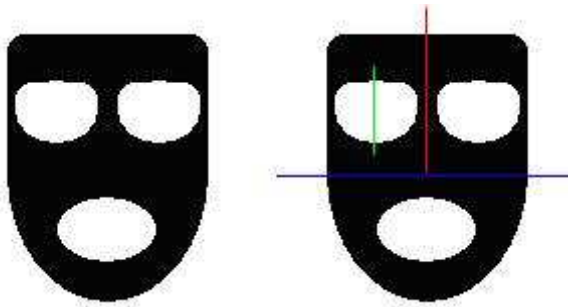

Figure 2.3: Template used in the face detection

difference between the gradient values of the eyes and the mouth holes, namely the white areas, and the black area of the template (as shown in figure 2.3) determines if that skin patch can be a face candidate. If the template matching's score is satisfactory then the algorithm moves to the next step. The template is designed in a way to return only edges in the areas where normal and non-rotated face of human eyes and mouth are most likely to be there.

The gradient values and edges are essential in our face detection algorithm. The value related to the edge scores directly influence the decision region. In our approach we are looking to find the best face match in the image. Therefore, among different face candidates the one with the highest face-score is being selected. The final score is calculated based on different factors such as symmetry of the gradient values between right and left eyes, net power which determines the gradient power captured in the eyes and mouth area and finally good/bad ratio that determines how high the difference between white and black area shown in figure 2.3 is. The number of edges that are captured and considered as the left eye,the right eye and the mouth and the edge-symmetry present in those areas determine the *Face Score*. These factors measure the face balance. Based on the *Face Score* value, the best face candidate will be nominated as the detected face.

2.2.2 Face Score

As explained above, each search box will be given a value that is specific to that section of the image, and this is called *Face Score* for that specific face candidate. The best face candidate then will be chosen among different face boxes based on the highest *Face Score*.

Since frontal faces are our main concern in this research, it is important to restrict the algorithm so it only captures and performs more sophisticated manipulations over the areas that are more probable in presence of a face. Therefore, as we follow the path to detect the face, the limits are more restricted to eliminate non frontal faces. First, skin area is specified so that the search continues to match the template over the skin area only. After matching the template, the net power and gradient values of the eyes and mouth are calculated and if they are symmetrical and greater than the values that are already found, this boxed area is considered as face. The search continues in the same manner until it finds the best case and the candidate.

Each search-box that is considered to be a face candidate is divided into 3 subsections as shown in figure 2.3. In the top section, based on the gradient values returned by the matched eyes location with template, the likelihood of eye location is found.

The symmetry of the eyes are also considered based on the alignments of the eye locations.The face-score is then updated based on the eyes symmetry. Also, mouth location is found using the highest gradient value present in the bottom section. The symmetry of the mouth box itself is being checked to achieve higher accuracy in finding a better face candidate.

In order to find different sized faces in the image, this algorithm is designed to find face candidates in different scale size images. In order to do this, the image is downscaled first and then kept fix for later processing. The template on the other hand shrinks to find different sized faces in the image. Figure 2.4 shows the search algorithm.

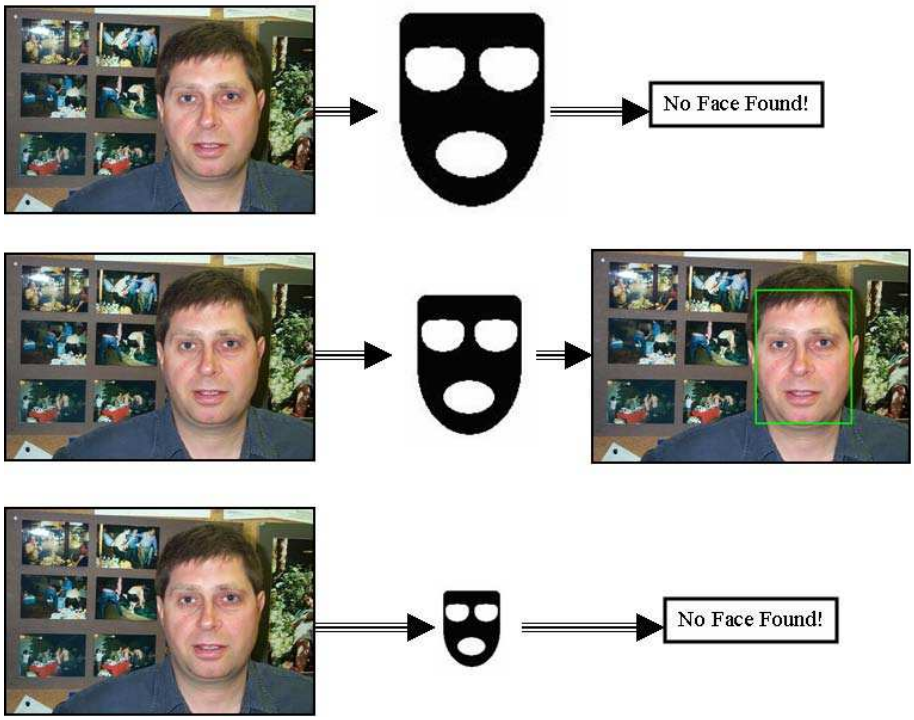

Figure 2.4: Searching In Different Size Modes

2.3 Comparison

For comparison purposes, the original system was tested in various stages of its completion and also with Conventional Neural Networks and EigenFace-based face detectors [40] A face detection experiment was performed on a set of 30 celebrity faces. These faces were mostly frontal without any rotation. Also, each image contained exactly one face. As a result, the reported results include only the detection rate, since ROC curves, number of false positives, and number of false negatives here are unnecessary. In essence, the number of false negatives (i.e. the missed faces) will be the same as the number of false positives (i.e. the incorrect face position estimates for the missed faces) and equal to 100 percent minus the detection rate [40]. The result of this test is given the table 2.1.

Original system results clearly shows high accuracy detection compared to the conventional methods, yet it keeps the simplicity and training efficiency over the other two

Face Detection Algorithm	Detection Rate in percent	RMSE for Correct Detections (in pixels)	RMSE for Incorrect Detections (in pixels)
EigenFace-Based	23.33	5.03	42.48
Conventional Neural Networks	86.67	8.00	23.23
Original System	93.33	4.96	69.38

Table 2.1: The correct face detection rates for various face detectors using a set of 30 celebrity images

methods. The eigenface method that is been presented here is the work done by [42] and has been used in several face detection and face recognition applications. This method forms a face subspace by calculating the eigenvectors from the face images in the training set [40]. Also, for conventional neural network system test, the neural network methodology of [43] was implemented for comparison with our novel fusion-based face detector. The neural network face detector takes an image of size 35x35 pixels as input and consists of a total of six layers. A shared weight neural network architecture such as the one described by [44] was utilized. The tagged images done by original system is shown in figure 2.5.

As previously discussed in this chapter, the primary machine was built for face detection purposes. The face detector that was developed uses different low-level analysis such as edge detection and skin detection to detect different face candidates and uses high-level analysis such as template matching to pin point the best face candidate. Since this system uses template matching as its base to detect faces, different problems associated with this type of method arise. A very common problem of templates is being rigid and uniform, which affects the detection rate. This is because faces are not all similar in

Figure 2.5: The zoomed in images of 30 celebrity faces used to test the various face detectors. The face detection results of the fused detector are shown on top of the images. Out of 30 images, only two detection errors (based on the face box coordinates) were made. The two errors are the rightmost two images in the bottom row.

shapes and sizes. In the next chapter we discuss the issues that are common in template matching algorithms, and introduce different methods to overcome these problems.

Chapter 3

Results

The system that was explained in Chapter 2 was built in Matlab for test purposes. The block diagram of figure 3.1 shows the system steps.

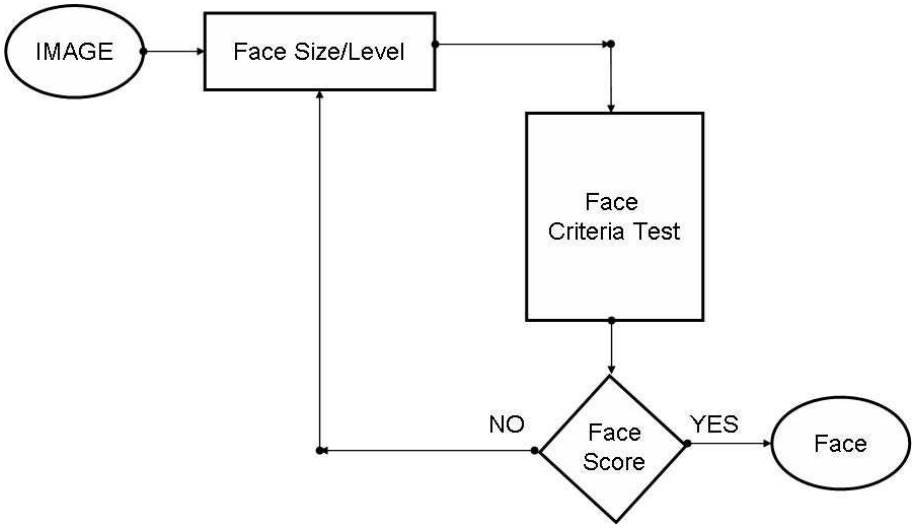

Figure 3.1: Original System's Block Diagram

The system applies the "Face size/level Query" on images. In this step the image is resized to the smaller picture to ease and fasten the face detection operation. Also since the edges play a significant role in finding faces in this algorithm, resizing is helpful to avoid discrepancies in face symmetry. The "Face Criteria Test" then detects faces, in the

rescaled image, based on skin detection, template matching and the overall face score. Although the initial algorithm shows a great performance of *91* percent detection rate but it misses faces in different occasions such as extreme face rotation. The performance of the original system is shown in figure 3.5.

Also template matching encounters many difficulties detecting a face location in the situations where different facial expressions are present or the face is experiencing rotation toward left or right. The examples in figure 3.2 show the system response in various situations.

To address the rotation issue, one might suggest the weighing method that was introduced by Krishnan Nallaperumal [8] to match the elliptical shape over the segmented skin area and find the rotation based on the segment's tilt. The orientation of the axis of the elongation determines the orientation of the region. The axis can be computed by finding the line for which the sum of the squared distance between region points and the line is a minimum. The angle of inclination is given by:

$$\theta = \frac{1}{2}\tan^{-}1(\frac{b}{(a-c)}) \tag{3.1}$$

where,

$$a = \sum_{i=1}^{n}\sum_{j=1}^{m} {x_{ij}}^2 B[i,j] \tag{3.2}$$

$$b = 2\sum_{i=1}^{n}\sum_{j=1}^{m} x_{ij}y_{ij}B[i,j] \tag{3.3}$$

$$c = \sum_{i=1}^{n}\sum_{j=1}^{m} {y_{ij}}^2 B[i,j] \tag{3.4}$$

$B[i,j]$ is the binary image information.

A negative point of this method is that the system highly relies on the skin detection technique. Generally skin detection depends on many different factors such as illumina-

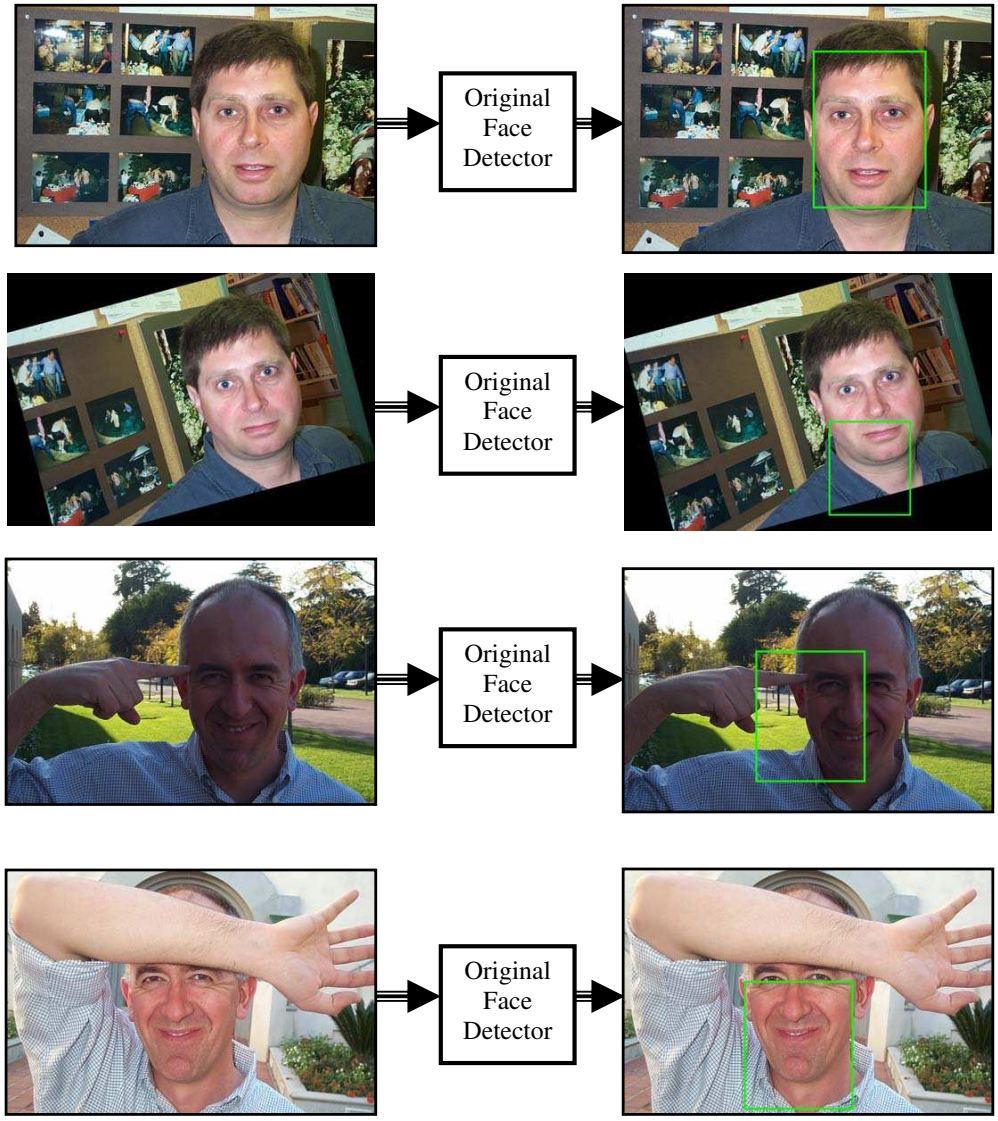

Figure 3.2: Failure examples of the original face detector

tion and lighting. Also background objects that have skin-like colours could be considered as skin segments. As shown in figures 2.2 and 2.1 the results of the skin detection are not very accurate and precise for face orientation detection.

Since skin detection on its own is not very reliable and losing any information at this primary stage will most likely result in false detection, the skin detection method with less restrict thresholds is used. In this case RGB skin detection method, as can be seen from the figures 2.2 and 2.1, returns a better and wider skin area. False skin areas will be eliminated in later stages with more accurate and precise parts of the face detection algorithm. In addition, making the skin detection threshold boundaries less strict, will affects the methods proposed by Krishnan Nallaperumal [8], since finding discrete oval patches of skin segments is not precise anymore.

Another suggestion for rotation detection is rotating the template over the skin segmented image to find the actual rotation angle. This idea is more promising since we are checking for the rotational angle exhaustively and the chance of missing a tilted face decreases extremely.

For rotation compensation, the system searches through the image exhaustively to detect the best face candidate in that rotation angle.It then rotates the whole image by some degrees and performs the face detection algorithm on the new angled image. If the new face candidate has a better face score than the previous one, this new face candidate will be the best face candidate for the time being. The search will continue in the same way until the best face candidate is chosen among all the face candidates returned in each rotation angle scenario. We have to keep in mind that our algorithm returns one face candidate per image with the highest *Face Score*. Since the image is rotated with different angles and search is performed over each rotated image, there remains several copies of the same image with different rotation angles. The final face candidate is chosen among all the face candidates returned in each rotation scene.

At this point one might ask rotating the image in every angle would give us the best

result in case of face detection, but how efficiently will the algorithm respond in this case. The Rotation resolution issue plays an important role in run-time and algorithm efficiency. Rotating the template or image every one degree is very promising in terms of finding the most suitable face candidate but it is not very efficient in terms of complex and expensive rotation calculation. Therefore, finding the best rotation angle is critical for the system performance where the correct face candidate is being detected while it has a descent running time.

3.1 Best Resolution Angle and Tilted Faces

To find the best and the most efficient rotation angle, the rotation block was added to the original system. The block diagram of the new system is shown in figure 3.3

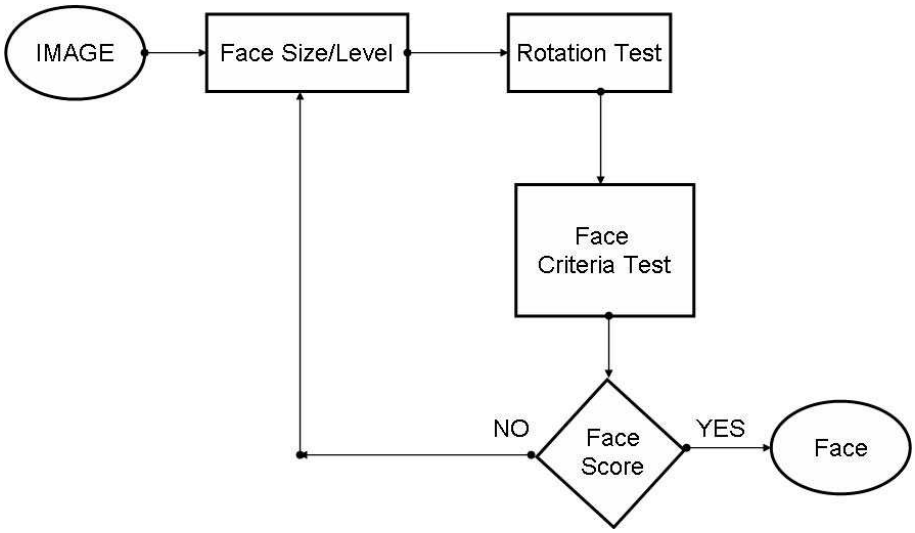

Figure 3.3: Block Diagram of the system with Rotation Block

The Algorithm that is introduced above was tested on Caltech University face data base which contains 450 color images with different faces and face expressions in different lighting conditions in complex background conditions with exactly one face existing in each image.

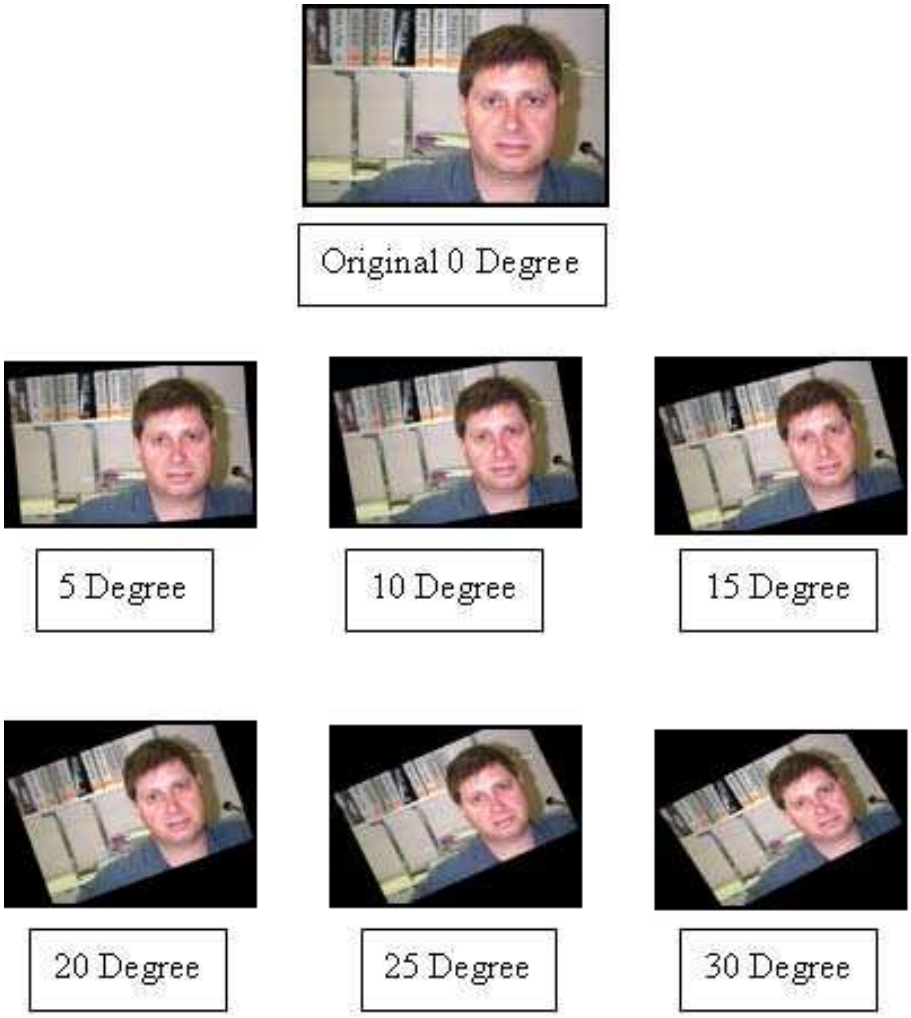

Figure 3.4: Rotational Database

The algorithm was tested on a variety of rotation angles that was manually introduced to the face database. The original database was assumed to be zero degree face rotation. So, original database was assumed to have straight up frontal faces. To produce different face angles to test our algorithm, the original data base was rotated every 5 degrees to form a new databases for that specific rotation angle. Then the Algorithm was tested on different databases to check its accuracy of face detection in different face angles. The original system's results is shown in figure 3.5. As can be seen from the graph, the original system is very dependent upon the face angles. In figure 3.5, the "rotation angles" axis refers to the specified angle database. For example, the 20 refers to the 20 degree rotated face database which contains 450 color images with exactly one face in it, where the original image is tilted by 20 degrees.

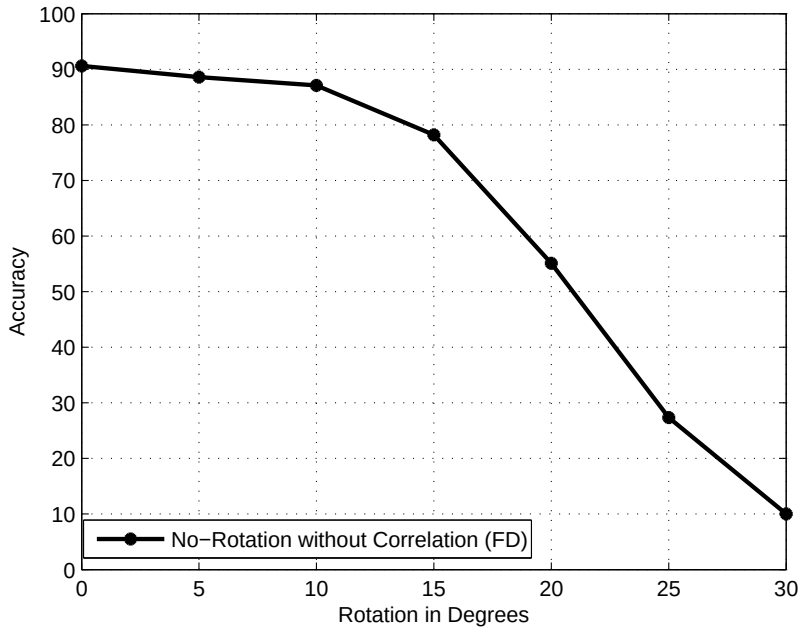

Figure 3.5: FD's Performance

A second test was performed in order to find the best rotation angle for our search. Rotating every 1 degree is very time consuming and even it might not be necessary to

perform; as such, finding the most efficient rotation angle is important. In order to find the best rotation angle for our rotation detection block shown in figure 3.3 various tests were needed to be performed on a variety of face-angle databases. These new databases were created from the original Caltech face database by rotating the original images by some degrees manually.

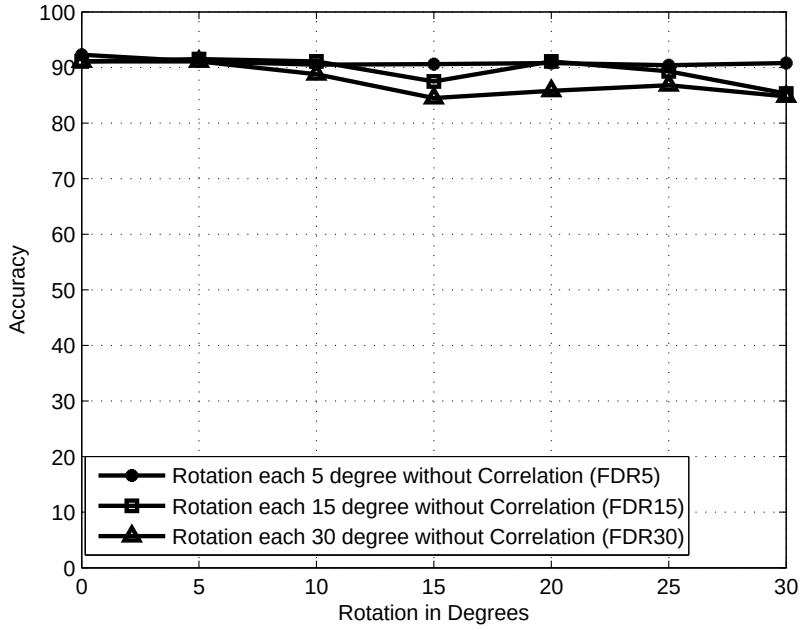

Figure 3.6: FDR5 vs. FDR15 vs. FDR30

To achieve an efficient system with high accuracy, the rotation feature was put to test in angles of 5, 15 and 30 degrees. Rotation feature adds the flexibility of rotating images to the algorithm, so that the system can find faces where in the first glance the original algorithm will not consider them as a typical frontal face configuration. Figure 3.6 shows the results of face detection with rotation feature added to the algorithm. Rotating each 5 degrees performs well, but since it is performing the search in each 5 degrees, the duration of the total search is long. Rotation angle of 15 degrees shows a very accurate and fast performance in comparison with the other test angles, and it can

be seen from the figure 3.6 that 15 degree of rotation gives almost the same accuracy in detecting frontal faces as the 5 degrees. As shown in figure 3.6, 15 degree rotation does not perform very well on 30 degree database and the reason is when the original images were tilted manually a black strip was introduced around the original images. These black strips introduce a solid edge to the system and also they make the images smaller than its original size when the system rotates the images backwards to make the frontal-face orientation.

Now, our algorithm can detect faces in different angles. The performance, as can be seen from the figure 3.6, shows a 90 percent accuracy in detecting faces. To increase this percentage, feature detection block was added to the existing machine.

3.2 Feature Scores And Face Score:

Now to boost the performance, a facial feature detector was added to the system. This block finds facial features inside the face candidate's search box and updates the face score accordingly. To test the performance of the system itself, the rotation detector was taken out for a fair test. The block diagram of the new system is shown in figure 3.7

To find the face feature, first the eyes and the mouth location was found using the most dominant edge score in the vicinity of the eye and the mouth location in the search box, and then correlation of the template and that specific eye or mouth area was calculated. Based on the correlation results face score was updated. Also after finding the exact location of the eyes and the mouth, based on feature balance in terms of location and placement of features in the normal up-right frontal face, face score was updated.

The following shows the mathematical derivation for template correlation. The discrete convolution of two functions $f(x, y)$ and $h(x, y)$ of size $M \times N$ is denoted by $f(x, y) * h(x, y)$ and is defined as the following expression:

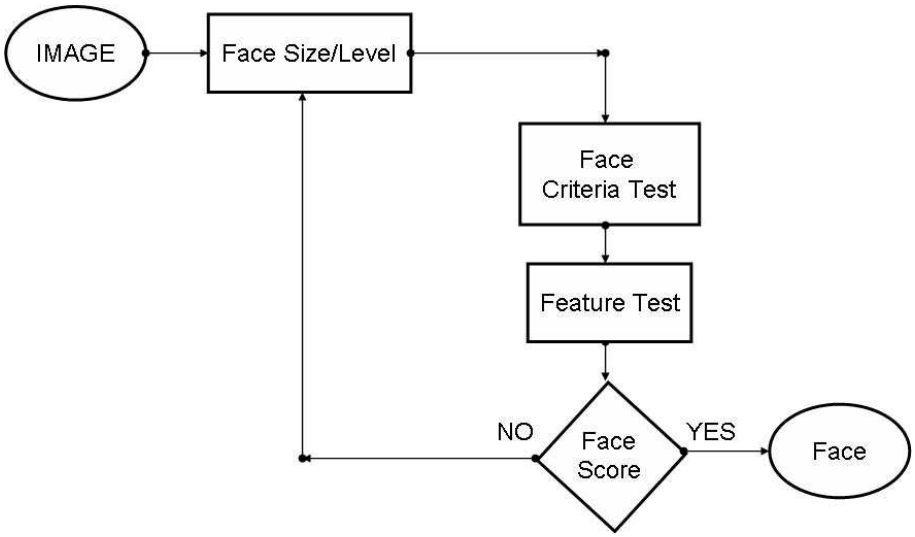

Figure 3.7: Block Diagram of the system with Feature detection block

$$f(x,y) * h(x,y) = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m,h)h(x-m,y-n) \tag{3.5}$$

The procedure of finding eyes and mouth is shown in figure 3.8.

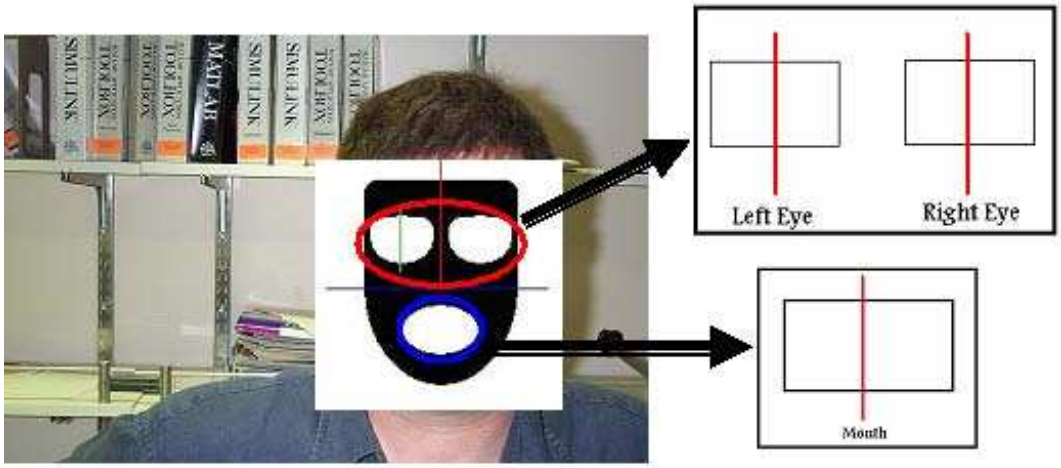

Figure 3.8: The templates search for the eye and the mouth location

For feature detection, simple eye and mouth templates were used. Eye and mouth templates that were used in feature matching are shown in the following diagrams, 3.9.

The issue here becomes the way we add the feature score to the face score that was already used in the original system.

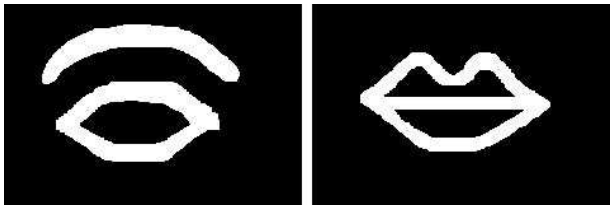

Figure 3.9: Eye and Mouth Templates

To obtain the feature scores from the eyes and mouth template matching first, edge detection was performed in the vicinity of the eyes and mouth location in the face candidate. Therefore, the search box was divided into 3 sections: top left, top right and the lower half face where mouth can be found. Thereafter, high gradient regions in each section were marked by little boxes and then the correlation of that specific region was tested with the eyes and mouth templates. The correlation results have a direct effect in face score computation. After finding the feature scores accordingly, the combination of the face score based on different features can be shown as following :

$$FaceScore = \frac{GBR \times NP \times SV \times pb1 \times pb2 \times pb3}{(1 + symmetrySum)} \tag{3.6}$$

Where in equation (3.6) the GBR is "goodBadRatio" and is derived from the ratio of the edge powers captured in the eyes and mouth area of the template known as "goodPower", and the edges captured in the black areas of the face template shown in figure 2.3 which are known as "badPower".

SV is the skin value of the search box on that specific location in the image. SV value is directly proportional to the amount of skin area captured in the search box.

The NP is the "netPower" and basically is the difference of the $goodPower$ and $badPower$ in that specific search box. $goodPower$ and $badPower$ derivations and equations are shown in 3.9,3.10. SV is the total skin power captured in the face candidate

area. The $pb1$, $pb2$ and $pb3$ are the balance scores related to the location of the eye and mouth features that are found in the search box. The symmetrySum is the total edge difference existing in the search box, therefore, if the search box has the perfect edge symmetry in vertical and horizantal direction, the symmetry score would be zero.

In order to find goodPower and badPower values the following sets of equations were used:

$$goodFace = FaceMask \times EdgesInTheSearchBox \tag{3.7}$$

where FaceMask has the value of one in the areas of eyes and mouth as shown in figure 2.3. To find the badFace portion of the face, we had to compliment FaceMask and the following equation was used:

$$badFace = (1 - FaceMask) \times EdgesInTheSearchBox \tag{3.8}$$

The goodPower and badPower are being calculated as following:

$$goodPower = \frac{\sum_{m=x}^{x+w} \sum_{n=y}^{y+h} goodFace}{\sum_{m=x}^{x+w} \sum_{n=y}^{y+h} FaceMask} \tag{3.9}$$

$$badPower = \frac{\sum_{m=x}^{x+w} \sum_{n=y}^{y+h} badFace}{\sum_{m=x}^{x+w} \sum_{n=y}^{y+h} FaceMask} \tag{3.10}$$

The netPower was caculated as following:

$$netPower = \max goodPower - badPower \tag{3.11}$$

The new system with the block diagram shown in figure 3.7 was tested on the Caltech Face Database. In this case, after finding the face candidate, the eye and the mouth templates were used to find the best the eye and the mouth location in the face candidate box. Based on the correlation results and also the placement of eyes and mouth in the

box, a new face Score was introduced to the system as explained above. The results of this scenario is shown in figure 3.10. As can be seen from the figure 3.10 the modified version performs better in harsher face angles. This can be promising when we are searching for faces with more than 15 degree of rotation.

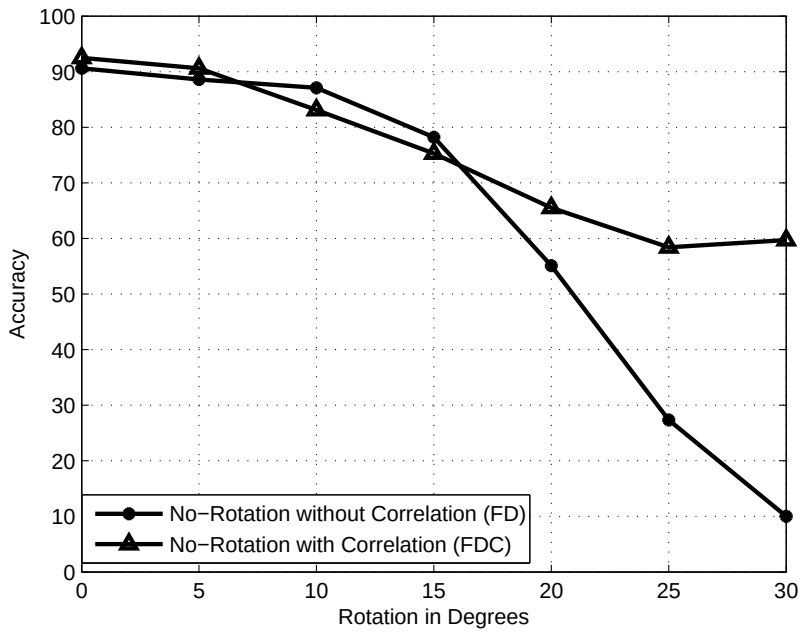

Figure 3.10: FD vs. FDC

As can be seen from figure 3.10, the feature detection increase the performance by 4 percent at frontal faces, and also it helps detecting the faces at the lower end with high tilted faces.

Now that the previous test implies the improvement of face detection with feature detection block, it is of interest to see how the combined blocks will performs together. Therefore, the two i.e. the rotation detection and the feature detection, blocks coexist to create the new face detection system to achieve higher accuracy and speed to detect faces in the images.

3.3 Complete Face Detection with Feature Criteria and Rotation detector

After combining all the different blocks that are developed so far, the new system block diagram looks as shown in the figure 3.11.

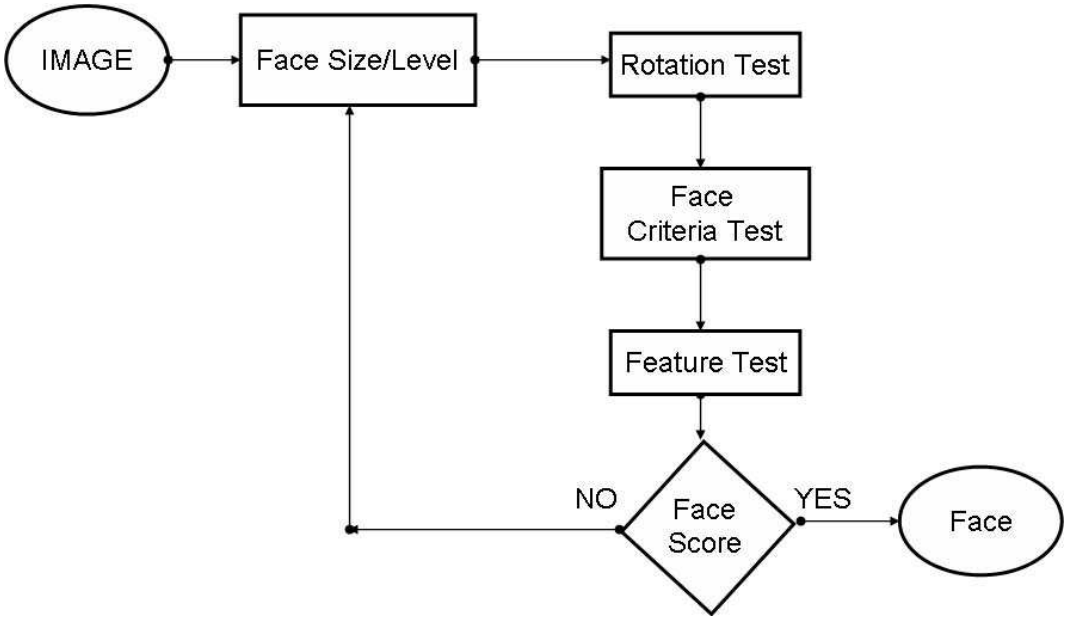

Figure 3.11: Block Diagram with Feature Criteria and Rotation Detection Blocks

Forth test was performed when feature extraction and rotation techniques are combined and added to the original face detection algorithm. In this case the performance of feature extraction at a set rotation of 30 degrees was tested. The system was tested with feature correlation namely "Face Detection Correlation Rotation 30" (FDCR30). Again the same system was tested under no correlation circumstances namely "Face Detection Rotation 30" (FDR30), which means the feature test block has been removed from the system for comparison purposes.

The test was done using Caltech University face database and the results are being shown in figure 3.12.

As can be seen from figure 3.12, The performance of the two graphs are very similar

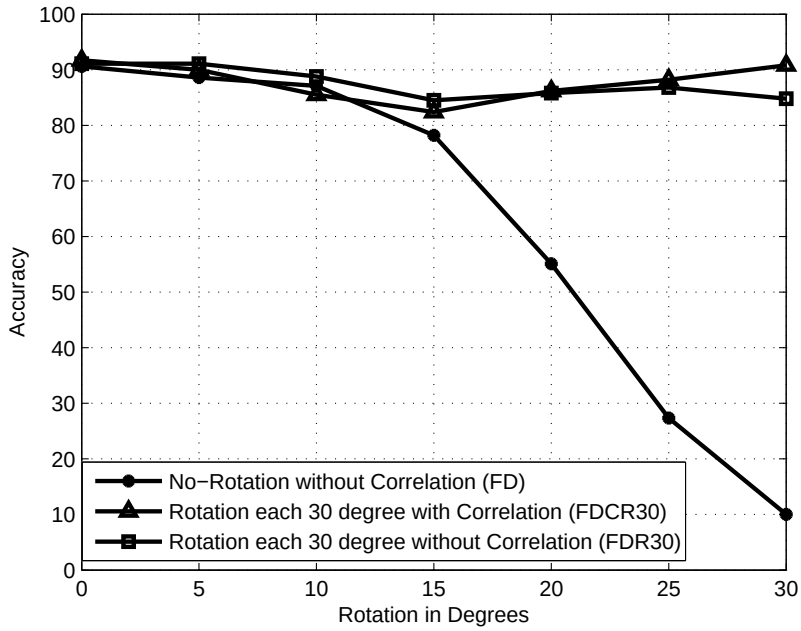

Figure 3.12: FD vs. FDCR30 vs. FDR30

and this could be predicted from figures 3.10. Since we have a uniform distribution of face angles and we have only 2 rotation angles of zero and 30 degrees, therefore, the system rotates the face if the face angle is more than 15 degrees. This results in having a 15 degree threshold bound. From figure 3.10 the performance result shows that although the feature correlation performs better at face angles close to zero and 30 degrees, they fail to respond properly to the faces with angles between 7.5 to 20 degrees. With later tests we will see the importance of the lower limit angle.

The results of the above tests suggest that changing the settings in rotation detection block helps increase the accuracy. As a new test case, the system has been tested with Rotation block set to 15 degrees.

The new test was done with feature correlation detection but in this case we introduced 15 degree rotation to our system. As can be seen from figure 3.13, the performance of face detector with feature correlation and 15 degree rotation (FDCR15) is better than

the same system but without feature correlation portion. In 15 degree rotation, the threshold bound moves to 7.5 degrees, and at the range of 0 to 7.5 degrees the correlation feature performs better, which could be predicted from figure 3.10.

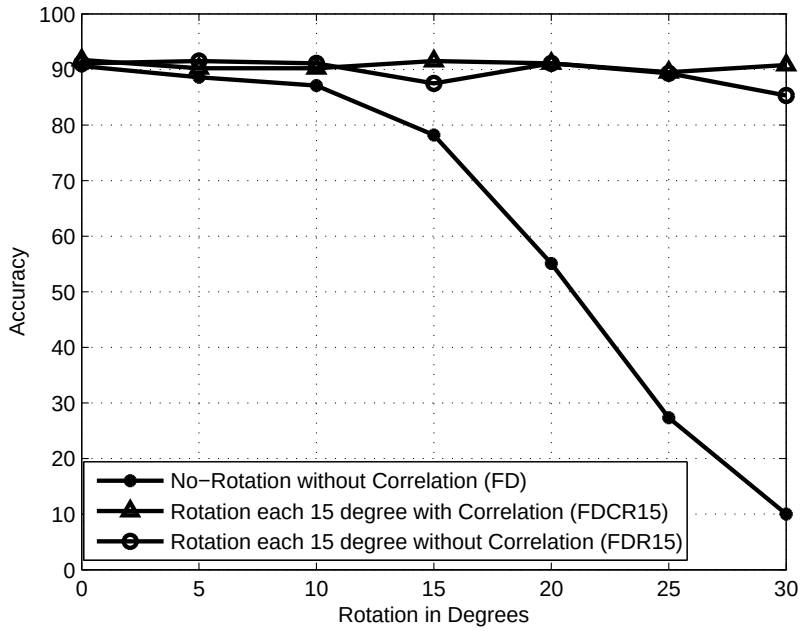

Figure 3.13: FD vs. FDCR15 vs. FDR15

As can be seen from the figure 3.13, 15 degrees of rotation with feature extraction portion improved the outcome significantly. Now examining the rotation resolution of each 5 degrees would be interesting. The results of this test are shown in figure 3.14.

It is obvious that the system's speed has been brought down by the choice of 5 degree rotation resolution, and as can be seen from figure 3.14,it does not increase the accuracy very much. Therefore choice of 15 degree rotation would be of interest for the optimal system.

Having seen the outcome of different cases of rotation and feature extractions, the best case of combination of the two different blocks would be with feature detection where rotation resolution is set to 15 degrees. The final graph is shown in figure 3.15.

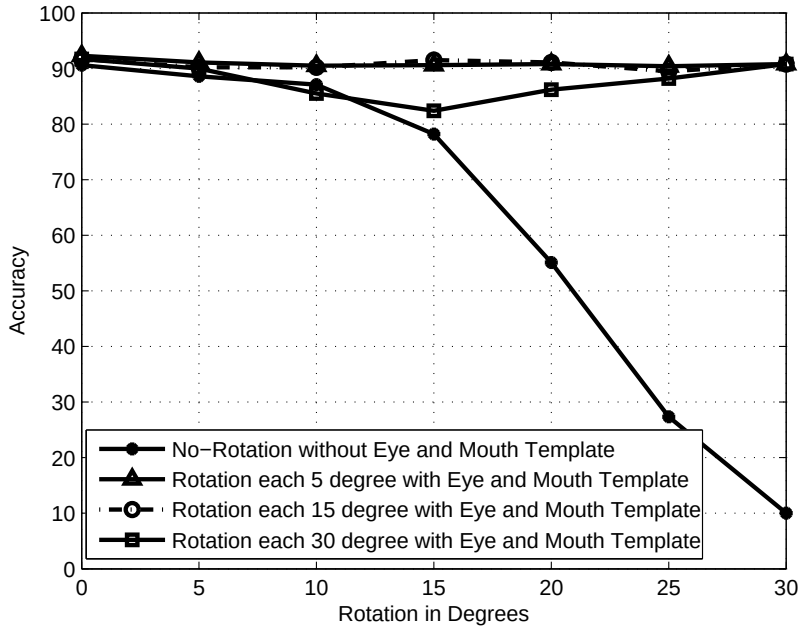

Figure 3.14: FD vs. FDCR5 vs. FDCR15 vs. FDCR30

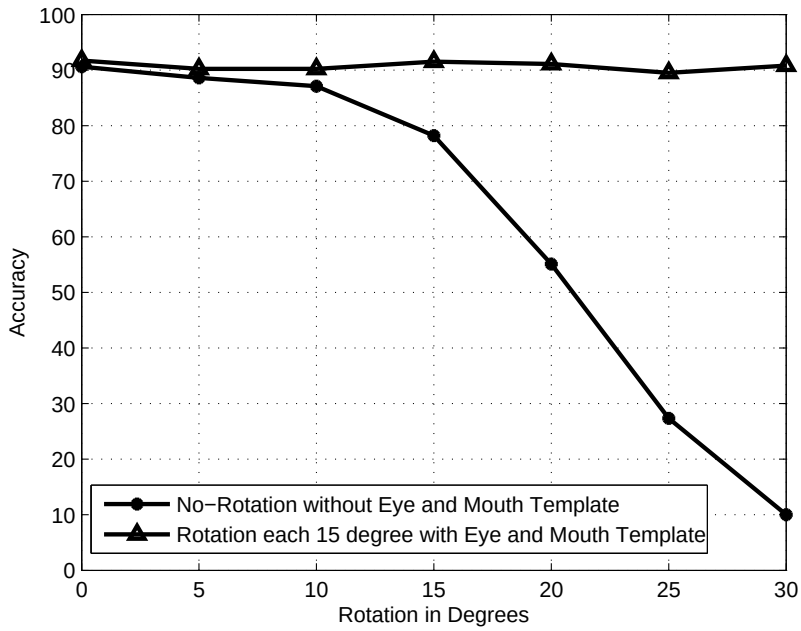

Figure 3.15: FD vs. FDCR15

Figure 3.15 clearly shows a huge improvement over the original face detector which was developed in chapter 2, and the side way rotation of faces do not change the performance of the new system. The overall improvement of 4 percent at zero angled frontal faces and a flat-line accuracy of 94 percent in different rotational angles show the improvements of this new system. The results of this new and enhanced face detector are shown in figure 3.16. Comparison between figures 3.2 and 3.16 demonstrates the enhancements achieved in the new system which is discussed in Chapter 3.

3.4 Comparison

After all the comparisons that are proved over the original system, for the last step, it was decided to test the enhanced machine and compare it with one of the very well known systems that are build over Viola and Jones method [41]. For that matter, OpenCV was chosen which is an open course computer vision library provided by Intel. OpenCV was tested over Caltech face database and the graph 3.17 was obtained from this test.

3.4.1 OpenCV first tag results

To achieve a fair comparison between our enhanced machine and OpenCV results, we first took the first tagged image returned by OpenCV. This decision was made since our system returns only one tag for an image where OpenCV returns multiple tags per image. So it would be only fair to take the first tagging returned by OpenCV. For this comparison the false positives are not taken into account and only images that has the first searched object detected as a face is taken into account. The results of this comparison is shown in figure 3.17.

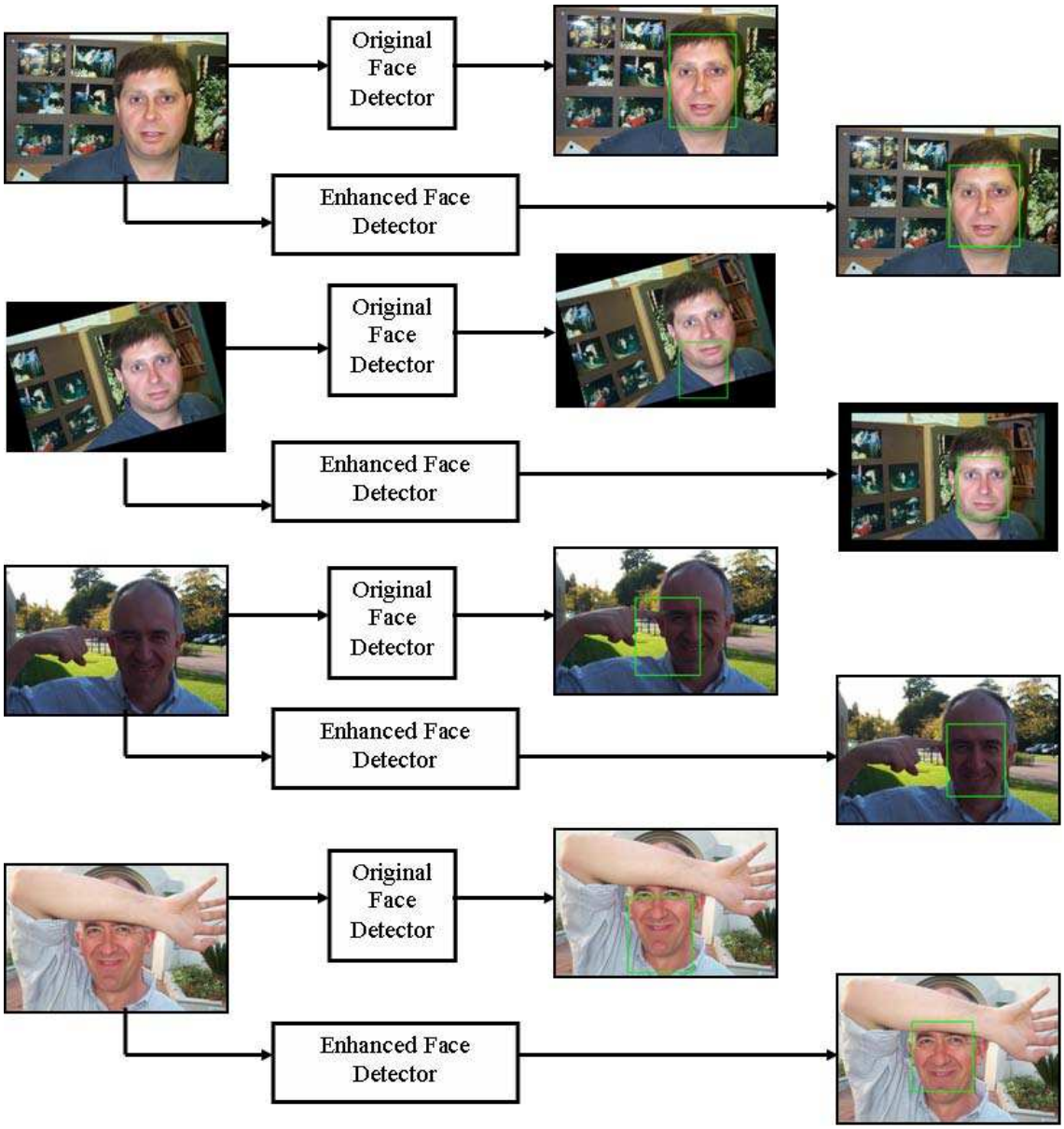

Figure 3.16: Results of the enhanced face detector vs. the original face detector

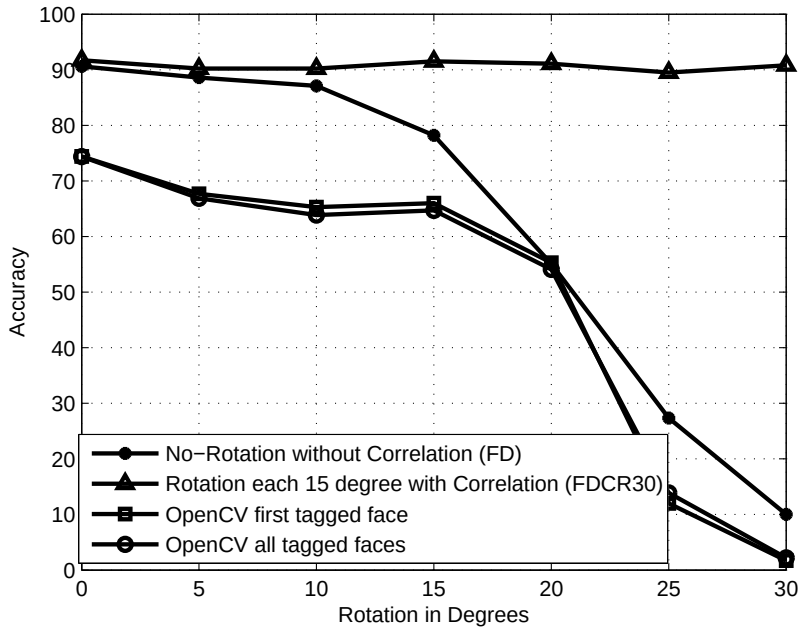

Figure 3.17: Results of the enhanced face detector vs. the original face detector

3.4.2 OpenCV all tag results

In this case we counted the number of false positives returned by OpenCV and false positives were calculated into accuracy result as shown in equation 3.12.

$$Accuracy = \frac{CD}{FP + T} \tag{3.12}$$

where CD is number of correctly tagged faces and FP is the number of wrong tagged objects and T stands for total number of images. Number of wrong tagged objects were set to zero in the case of first tag taken from OpenCV.

From the figure 3.17, it can be seen that our system response is very robust. Although OpenCV has a decent detection rate, but the false detections bring the accuracy of the system as low as 70 percent.

3.4.3 OpenCV best tag results

OpenCV has very high accuracy of detection, although at the same time this method returns many false positives. For that matter, We decided to discard any false detection done by OpenCV and only count the number of objects that are returned accordingly. The results of this comparison versus our enhanced system is shown in figure 3.18.

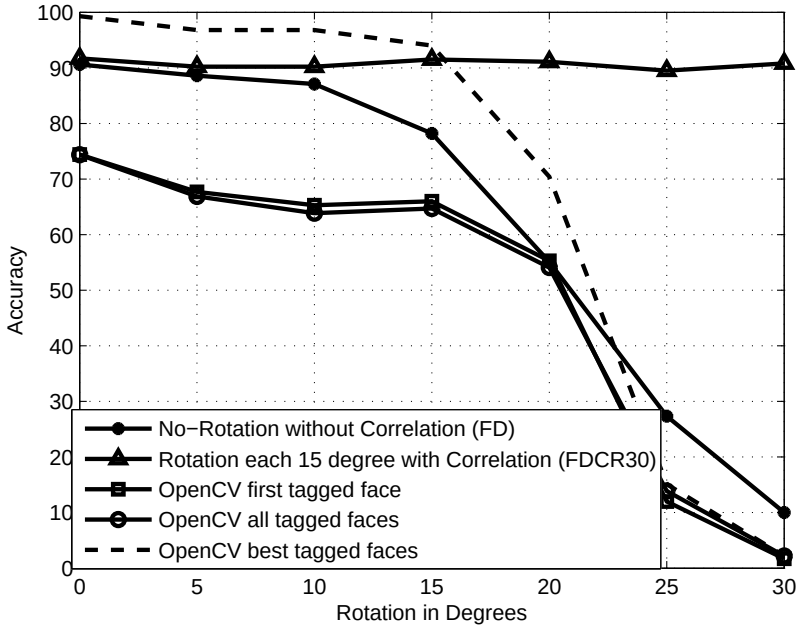

Figure 3.18: Results of the enhanced face detector vs. OpenCV

Again the results in figure 3.18 shows the better performance of our system tested on Caltech Face database compared to OpenCV provided by Intel. Our enhanced system keeps its invariance toward rotation and shows the detection rate of 94 percent. Also it is interesting to mention that our system does not need any training to achieve this high accuracy and yet this is another reason for our simple system to be taken as a more efficient face detector compared to OpenCV. Although when OpenCV's best result shows high detection rate, but still it does not contain its accuracy over all spectrum of rotated faces (we have to mention here again that for OpenCV's best result, false positives are

not taken into account).

3.5 Summary

In chapter 3, based on the original face detector and the two components that were added to the system, the overall performance and accuracy of the face detector was improved. First, by addition of Rotation Detection Block, the system was able to detect non-frontal faces that were tilted toward left or right. This block improved the accuracy of the system in detecting different faces with different angles, therefore the system became more robust and independent of the face angles. Then to improve the accuracy of the system, the feature detection blocks were added. This block which works based on template matching criteria, boosted the performance of the system. The combination of the two blocks created a system with high accuracy in detecting different angled faces in the images. The final system's performance is shown in figure 3.15. Also the comparison results are shown in figure 3.17 where it further emphasizes on the improvements that are done on the original system block.

Chapter 4

Conclusion

The problem of different face poses and angles is one of the key challenges in the area of face detection. Different approaches in face detection have taken different paths toward solving this problem. Furthermore, the main concern in feature-based approaches is dealing with varying objects. In our case, where a rigid template is used to detect the object of interest in a complex background, the main concern was faces.

4.1 Conclusion

All face detection algorithms can be categorized in two main groups of feature-based and Model-based approaches. The weakness of Model-based approach is the large number of training that has to be performed on the system for faces and non faces, so that the system can detect faces in different background complexities. In addition, the training in most of these systems are carried out off-line. Also, Model-based approaches are slow and creating a real-time system using this method is less feasible. On the other hand, Feature-based approaches are fast and can work real-time with minimal training. The downfall of this method is inaccuracy. For this reason we tried to make a better system using this feature-based approach which has all the advantages of feature-based machine and also is very accurate.

The two main problems of feature-based approach and specially Template matching algorithm are first using a rigid and strict face model to detect all face objects with different posses and face expressions. The second problem is using the template to detect tilted faces where matching the template can not be accomplished because of the template orientation.

In most template matching problems, correlation of the template and faces is considered to be the criteria of matching; whereas in our case, we used face template to determine the symmetry of face objects and detect the face based on the high edge concentration regions which could look like a face feature such as the eyes and the mouth. Using the template in such a unique way made our approach overcome the rigidity of face templates.

To overcome the orientation problem introduced to everyday photos taken by ordinary people, there has been different proposals. One method that was introduced in Chapter 2 used the weight and orientation of skin segmented image to achieve the face angle which was very interesting but done poorly because of skin segmentation problems. In our approach, we have used the rotating image procedure where the system rotates the image exhaustively to find the best matching face among all the situations. This approach is very time consuming and expensive, therefore the best rotation angle was found by performing different test cases on Caltech Face Database. This rotating feature which gives the best performance and accuracy has the rotation angle of 15 degrees.

To extend our work and boost the accuracy of our face detector, fine-tuned feature detector was added to the system. In this case, eye and mouth templates were used to correlate with the eye and the mouth locations in the face candidate and their correlation scores were counted toward determining the best face case. Since a very simple eye and mouth template were used to correlate with the edge version of the face, a great improvement was not achieved in this case, but more research in this area is necessary to fine tune the feature detection.

In summary, the comparison that is been done among the primary work and other methods and also the comparison of the enhanced system and the primary work and OpenCV shows the enhancements achieved over the original system, and therefore, all the systems that are being compared with the original system. With this research we have attained a robust system that does not require any sort of training and yet can detect faces in images very accurately and efficiently.

4.2 Future Work

- Most of the face detectors use skin detection as their primary module toward decreasing the search area and increasing the accuracy of searched results. Since it is very critical to avoid any dismissal of faces at the beginning, a very good skin segmentation can help both the accuracy and performance.

- The system that was developed was tested on databases with one face existing in the image. This work can be extended to detect multiple faces in the picture. Also defining a better lower threshold for faceScore can avoid marking any non faces in the photos with no faces existing in them.

- Feature detection as discussed in the conclusion section, can be improved to achieve higher accuracy of face features and consequently reaching a better face detector. A more sophisticated eye and mouth template can be very helpful toward this goal.

Bibliography

[1] R. Brunelli and T. Poggio, Face recognition: Feature versus templates, *IEEE Trans. Pattern Anal. Mach. Intell., pp. 1042-1052, 1993.*

[2] J. Choi, S. Kim, and P. Rhee, Facial components segmentation for extracting facial feature, in *Proceedings Second International Conference on Audio- and Video-based Biometric Person Authentication (AVBPA),March 1999.*

[3] T. Kawaguchi, D. Hidaka, and M. Rizon, Robust extraction of eyes from face, in *Proceedings of the 15th International Conference on Pattern Recognition, Vol. I, pp. 1109-1114, 2000.*

[4] J. L. Crowley and F. Berard, Multi-modal tracking of faces for video communications, in *IEEE Proc. IEEE Conf. on Computer Vision and Pattern Recognition, Puerto Rico,pp. 640-645, Jun. 1997.*

[5] S. Kawato and J. Ohya, Real-time detection of nodding and head-shaking by directly detecting and tracking the between-eyes in *Proceedings Fourth IEEE International Conference on Automatic Face and Gesture Recognition, pp. 40-45, 2000.*

[6] Q. B. Sun, W. M. Huang, and J. K. Wu, Face detection based on color and local symmetry information, in *IEEE Proc. of 3rd Int. Conf. on Automatic Face and Gesture Recognition,pp. 130-135, 1998.*

[7] K. Yachi, T. Wada, and T. Matsuyama, Human head tracking using adaptive appearance models with a fixed-viewpoint pan-tilt-zoom camera, in *Proceedings Fourth IEEE International Conference on Automatic Face and Gesture Recognition, pp 150-155, 2000.*

[8] K.; Subban, R.; Krishnaveni, K.; Fred, L.; Selvakumar, R.K.;Human face detection in color images using skin color and template matching models for multimedia on the Web Nallaperumal, *Wireless and Optical Communications Networks, 2006 IFIP International Conference on 11-13 April 2006, pp. 5-10*

[9] Sanjay Kr. Singh, D. S. Chauhan, Mayank Vatsa, Richa Singh, *A robust skin color based face detection algorithm, Tamkang Journal of Science and Engineering*, vol. 6, no. 4, pp. 227-234, 2003.

[10] Rein-Lien Hsu, M. Abdel-Mottaleb, A. K. Jain, *Face detection in color images, IEEE Trans.* PAMI , vol. 24, no. 5, pp. 696-706, 2002.

[11] K. Fukunaga, *Introduction to statistical pattern recognition*, Boston: Academic Press, 2nd edition,1990.

[12] D. Chai and A. Bouzerdoum, *A Bayesian Approach to Skin Color Classification in YCbCr Color Space*, In Proceedings IEEE Region Ten Conference (TENCON2000), vol. 2, pp. 421-424.

[13] PEER, P., KOVAC, J., AND SOLINA,Human skin colour clustering for face detection. *In International Conference on Computer as a Tool. EUROCON2003, 2003.*

[14] AHLBERG, *A system for face localization and facial feature extraction* Tech. Rep. LiTH-ISY-R-2172, Linkoping University., July. 1999.

[15] R. Chellappa, C. L.Wilson, and S. Sirohey. *Human and machine recognition of faces: A survey*, Proc. IEEE, vol. 83, no. 5, pp. 705-740, 1995.

[16] T. Sakai, M. Nagao, and T. Kanade, Computer analysis and classification of photographs of human faces,*presented at the 1st USAJapan Comput. Conf.. Session 2-7-1. 1972*

[17] D. Valentin, H. Abdi, A. J. OToole, and G. Cottrell, Connectionist models of face processing: *A survey,Pattern Recog. 27, 1994, 1209-1230.*

[18] HJELMAS, E. AND LOW, B. K. *Face detection:A Survey.* Comput. Vis. Image Understand. 2001., pp. 236-274.

[19] P. J. L. Van Beek, M. J. T. Reinders, B. Sankur, and J. C. A. Van Der Lubbe, *Semantic segmentation of videophone image sequences*, in Proc. of SPIE Int. Conf. on Visual Communications and Image Processing, 1992, pp. 1182-1193.

[20] J. L. Crowley and F. Berard, *Multi-modal tracking of faces for video communications*, Proc. IEEE Conf. Computer Vision and Pattern Recognition, pp. 640-645, 1997.

[21] E. Hjelmas and J. Wroldsen, *Recognizing faces from the eyes only*, in Proceedings of the 11th Scandinavian Conference on Image Analysis, 1999.

[22] HSU, R.-L., ABDEL-MOTTALEB, M., AND JAIN, A. K. *Face detection in color images.* IEEE Trans. Pattern Analysis and Machine Intelligence 24, 5, pp. 696-706, 2002.

[23] S. H. Jeng, H. Y. M. Liao, C. C. Han, M. Y. Chern, and Y. T. Liu, *Facial feature detection using geometrical face model: An efficient approach, Pattern Recog. 31, pp. 273-282., 1998.*

[24] M. C. Burl, T. K. Leung, and P. Perona, *Face localization via shape statistics*, in Int. Workshop on Automatic Face and Gesture Recognition, Zurich, Switzerland, pp. 154-159, June 1995.

[25] W. Huang and R. Mariani, *Face detection and precise eyes location*, in Proceedings of the 15th International Conference on Pattern Recognition, 2000, Vol. 4, pp. 722-727.

[26] C. Lin and K. Fan, *Human face detection using geometric triangle relationship*, in Proceedings of the 15th International Conference on Pattern Recognition, 2000, Vol. II, pp. 941-944.

[27] M. Kass, A. Witkin, and D. Terzopoulos, *Snakes: active contour models*, in Proc. of 1st Int Conf. on Computer Vision, London, 1987. pp. 259-268.

[28] H. Wang, S. F. Chang. *A highly efficient system for automatic face region detection in MPEG video*, IEEE trans. Circuit Sys. for Video Tech., vol. 7, no. 4, pp. 615-628, 1997.

[29] A. L. Yuille, P. W. Hallinan, and D. S. Cohen, *Feature extraction from faces using deformable templates*,Int. J. Comput. Vision 8, 1992, 99-111.

[30] T. F. Cootes and C. J. Taylor, *Active shape modelssmart snakes*, in Proc. of British Machine Vision Conference, 1992, pp. 266-275.

[31] E. Viennet and F. Fogelman Soulie, *Connectionist methods for human face processing, in Face Recognition: From Theory to Application.* Springer-Verlag, Berlin/New York, pp. 124-156, 1998.

[32] S.-H. Lin, S.-Y. Kung, and L.-J. Lin, *Face recognition/detection by probabilistic decision-based neural network*, IEEE Trans. Neural Networks 8, 1997, pp. 114-132.

[33] Rafael C. Gonzalez, and Richard E. Woods. (2002). chapter:Image Enhancement in Spatial Domain, *Digital Image Processing*. Publisher: Tom Robines.

[34] C. Garcia, G. Tziritas. *Face detection using quantized skin color regions merging and wavelet packet analysis*, IEEE Trans. Multimedia, vol. 1, no. 3, pp. 264-277, 1999.

[35] Y. Guan, P. Wu, N. Lu, X. Liu, *Efficient and robust face detection from color images*; Proceedings of the 6th World Congress on Intelligent Control and Automation, June 21 - 23, 2006,, pp. 9664- 9668

[36] J. Wang, Y. Gu, K.N. Plataniotis, A.N. Venetsanopoulos,*Select eigenfaces for face recognition with one training sample per subject* Control, Automation, Robotics and Vision Conference, ICARCV 2004 8th; Vol. 1, pp. 391-396

[37] J. Wang, K.N. Plataniotis, A.N. Venetsanopolous,*Combining features and decisions for face detection* Acoustics, Speech, and Signal Processing, 2004. Proceedings. (ICASSP '04). IEEE International Conference on 2004; Vol. 5, pp. 717-720

[38] D Nguyen, D Halupka, P Aarabi, A Sheikholeslami, *Real-Time Face Detection and Lip Feature Extraction Using Field-Programmable Gate Arrays*; IEEE Transactions on Systems, Man, and Cybernetics, Part B, Vol. 36, No. 4, pp. 902-912, August 2006.

[39] SA Rabi, P Aarabi, *Face Fusion: An Automatic Method For Virtual Plastic Surgery*; Proceedings of the 2006 International Conference on Information Fusion (Fusion'06), Florence, Italy, July 2006.

[40] P. Aarabi, P. Lam, A. Keshavarz,*Face detection using information fusion*; International Conference on Information Fusion, 2007, pp. 1-8

[41] Viola and Jones, *Robust Real-Time Face Detection*; International Journal of Computer Vision, 2004, pp. 137-154

[42] Turk, M. and Pentland, *Eigenfaces for recognition*; The Journal of Cognitive Neuroscience,A. 1991, pp. 71-86

[43] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, *Gradient-Based Learning Applied to Document Recognition*; Proc. IEEE, vol. 86, no. 11, pp. 2278-2324, 1998

[44] Vaillant, R., Monrocq, C., and LeCun, Y., *Original approach for the localization of objects in images*; IEEE Proceedings on Vision, Image, and Signal Processing, vol. 141, pp. 245-250, 1994.